Remove logarithmic factor in competitive ratio for reusable-resource Bayesian online assortment

ChatGPT (`gpt-5.6-sol`)*

Partial progress generated on 2026-08-24

Abstract

This remains partial progress, not a complete solution. The arbitrary-duration $1 - O(c_{\min}^{-1/2})$ gate theorem and its polynomial-time implementation remain unproved. The corrected results below do, however, repair the boundary cases, the computational claims, and the scope of the recurring-chain theorem.

1 The open problem

This paper addresses an open problem posed by Yiding Feng, Rad Niazadeh, Amin Saberi in *Near-Optimal Bayesian Online Assortment of Reusable Resources* [1] (Economics and Computation (EC), 2022), Section 4 (Conclusion), page 21. The website catalogue entry is problem 24 (`W4285091035_p0`) of the Open Problems in Operations Research collection.

> *Determine whether, for the Bayesian online assortment problem with reusable resources and general rental-duration distributions, the logarithmic factor in the known large-inventory competitive ratio can be removed.*
>
> *Concretely: is there a polynomial-time online algorithm whose competitive ratio against* Expected-LP$[\{F_t\}]$ *scales as*
>
> $$1 - O\left(\frac{1}{\sqrt{c_{\min}}}\right)$$
>
> *in the large-inventory regime (i.e., as $c_{\min} \to \infty$), improving over the $1 - O(\sqrt{\log(c_{\min})/c_{\min}})$ rate currently obtained for general rental-duration distributions?*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

---

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.10 repair write-up; run `run_20260824_074958_af77603b, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

2 Constructive reduction from assortments to interval gates

Let y^* solve the Bayesian Expected-LP, and define

$$q_{i,t}^z = \sum_{S \in \mathcal{S}} \phi^z(S,i)\, y^*_{S,t,z}.$$

Then

$$\sum_{\tau \le t,\, z} F_\tau(z)\, q_{i,\tau}^z \, \Pr(D_i^z > t-\tau) \le c_i. \tag{1}$$

The gate-balance requirement below is *ex ante*,

$$\Pr(G_{i,t} = 1 \mid Z_t = z) \ge \gamma_i, \tag{2}$$

rather than conditional on every past history; history-conditional balance is generally impossible whenever a full-capacity history has positive probability.

The required assortment conversion rests on the following lemma.

Lemma 1 (Masking)**.** *Let $B \in \mathcal{S}$, where $B = \varnothing$ is allowed, and suppose that*

$$0 \le x_i \le \phi^z(B,i) \quad (i \in B), \qquad x_i = 0 \quad (i \notin B).$$

Then there is a lottery over subsets of B whose choice-probability marginals equal x.

Proof. The assertion is immediate for $B = \varnothing$. If some $k \in B$ satisfies $\phi^z(B,k) = 0$, then $x_k = 0$, and weak substitutability gives

$$x_i \le \phi^z(B,i) \le \phi^z(B \setminus \{k\}, i).$$

Delete k and apply induction.

Otherwise every $\phi^z(B,i) > 0$. Put

$$\lambda = \min_{i \in B} \frac{x_i}{\phi^z(B,i)}$$

and choose a minimizer k. If $\lambda = 1$, then $x_i = \phi^z(B,i)$ for all i, so offer B. If $\lambda < 1$, set

$$x_i' = \frac{x_i - \lambda \phi^z(B,i)}{1-\lambda} \quad (i \ne k), \qquad x_k' = 0.$$

Then

$$0 \le x_i' \le \phi^z(B,i) \le \phi^z(B \setminus \{k\}, i).$$

Mix B, with probability λ, with the inductively obtained lottery on $B \setminus \{k\}$. □

At time t, after observing z, sample all product gates before sampling the LP assortment or the current choice. Let E be the set of open gates. Independently sample S with probabilities $y^*_{S,t,z}$, assigning the residual mass to $\varnothing$. Put $B = S \cap E$ and apply Lemma 1 with

$$x_i = \phi^z(S,i)\, \mathbf{1}\{i \in B\}.$$

Since $B \subseteq S$,

$$\phi^z(S,i) \le \phi^z(B,i).$$

Consequently,

$$\Pr(i \text{ is chosen} \mid \text{past}, z, E) = q_{i,t}^{z}\, \mathbf{1}\{i \in E\}. \tag{3}$$

For each product, the actual choice on an open gate, together with an independently simulated choice on a closed gate, forms the required latent Bernoulli sequence. Hence

$$\begin{aligned} \mathbb{E}[\text{Revenue}] &= \sum_{t,z,i} F_t(z)\, r_i^z\, q_{i,t}^z\, \Pr(G_{i,t} = 1 \mid Z_t = z) \\ &\geq \min_i \gamma_i \,\text{Expected-LP}. \end{aligned} \tag{4}$$

Thus a polynomial-time interval-gate algorithm would yield the desired polynomial-time assortment algorithm: the Expected-LP is handled by the assumed assortment oracle, and the masking lottery has support of size at most $|S| + 1$.

The converse separation remains only information-theoretic. For a single product, let P be the compact convex set of achievable service vectors

$$s_{t,z} = \Pr(Z_t = z,\, A_t = 1,\, G_t = 1), \qquad a_{t,z} = F_t(z) \Pr(A_t = 1 \mid Z_t = z).$$

The set P is downward closed, by virtual simulation followed by coordinatewise suppression. If P contains no $s \geq \gamma a$, strong separation from $\gamma a + \mathbb{R}_+^m$ produces $w \geq 0$ satisfying

$$\sup_{s \in P} w \cdot s < \gamma\, w \cdot a. \tag{5}$$

Using $w_{t,z}$ as rewards gives a one-product instance whose Expected-LP is $w \cdot a$. This proves information-theoretic equivalence, but it does not produce a polynomial-time gate policy. An exact scenario-tree occupation-measure LP exists, but its configuration state space is exponential in T and c.

3 Rigorous general interval bounds

Let Γ_c denote the worst-instance optimal ex-ante balanced gate factor for finite independent stochastic intervals of pointwise expected depth at most c.

3.1 Universal $1/2$ existence result

For every integer $c \geq 1$,

$$\Gamma_c \geq \frac{1}{2}. \tag{6}$$

Indeed, inductively require every earlier gate to have marginal exactly $1/2$. If $H_t \leq c$ denotes the accepted occupancy before time t, then gate-before-outcome independence gives

$$\mathbb{E} H_t = \frac{1}{2} \sum_{s<t} \Pr(A_s = 1,\, D_s > t - s) \leq \frac{c}{2}.$$

Therefore

$$\Pr(H_t = c) \leq \frac{\mathbb{E} H_t}{c} \leq \frac{1}{2},$$

so probability mass at least $1/2$ lies on nonfull histories. Randomizing over those histories produces a feasible gate with marginal exactly $1/2$.

This is an existence construction; computing the required history probabilities exactly may take exponential time. For $c = 1$ the bound is sharp. Take two permanent requests with activation probabilities $1 - \varepsilon$ and ε. If both gate marginals are at least γ, then the second gate can open only when the first request was not accepted, so

$$\gamma \leq 1 - (1 - \varepsilon)\gamma, \qquad \gamma \leq \frac{1}{2 - \varepsilon}.$$

Letting $\varepsilon \downarrow 0$ gives $\Gamma_1 = 1/2$.

3.2 Polynomial-time logarithmic guarantee

For $c \geq 9$, put

$$\delta = 2\sqrt{\frac{\log c}{c}} < 1.$$

Independently mark every tentative request with probability $1 - \delta$, and open its gate if and only if it is marked and physical capacity is available.

Before time t, define the marked infinite-server occupancy

$$Y_t = \sum_{s<t} B_s A_s \, \mathbf{1}\{D_s > t - s\}.$$

The physical occupancy is at most Y_t pathwise, while

$$\mu_t = \mathbb{E}Y_t \leq (1 - \delta)c.$$

The summands are independent across s, even when A_s and D_s are correlated within a request. The Poisson-binomial Chernoff bound

$$\Pr(Y \geq \mu + a) \leq \exp\left(-\frac{a^2}{2\mu + a}\right)$$

therefore gives

$$\Pr(Y_t \geq c) \leq \exp\left(-\frac{\delta^2 c}{2 - \delta}\right) \leq c^{-2}.$$

Hence

$$\Gamma_c \geq \left(1 - 2\sqrt{\frac{\log c}{c}}\right)(1 - c^{-2}). \tag{7}$$

3.3 Necessary $c^{-1/2}$ loss

Take n requests with deterministic duration beyond the horizon and

$$A_t \sim \mathrm{Ber}(c/n),$$

independently. The pointwise expected depth is at most c. If every gate has probability at least γ, gate-before-activation independence implies

$$\mathbb{E}[\#\ \text{accepted}] \geq \gamma c.$$

Writing $X_n \sim \mathrm{Bin}(n, c/n)$, feasibility gives

$$\gamma c \le \mathbb{E}\min(X_n, c).$$

Letting $n \to \infty$, $X_n \Rightarrow X \sim \mathrm{Pois}(c)$, and

$$\mathbb{E}(X-c)^+ = c\Pr(X=c).$$

Thus

$$\Gamma_c \le 1 - \Pr(\mathrm{Pois}(c)=c) = 1 - \frac{e^{-c}c^c}{c!} = 1 - \frac{1}{\sqrt{2\pi c}} + O(c^{-3/2}). \tag{8}$$

Combining (6)–(8), for $c \ge 9$,

$$\boxed{\max\left\{\frac{1}{2}, \left(1 - 2\sqrt{\frac{\log c}{c}}\right)(1-c^{-2})\right\} \le \Gamma_c \le 1 - \frac{1}{\sqrt{2\pi c}} + O(c^{-3/2}).} \tag{9}$$

The upper and lower losses still differ by a factor of $\sqrt{\log c}$.

4 Corrected fixed recurring-chain theorem

Let $c \ge 1$, and let E be a finite set. For each $e \in E$, assume the following:

- $x_e \in (0,1]$ is fixed across all epochs;
- $\sum_e x_e \le c$;
- the epochs of e are nonoverlapping;
- the complete renewal schedule is exogenous and independent of the initial shadow sample, the activations, and the gate coins;
- activations at renewals are fresh $\mathrm{Ber}(x_e)$ variables;
- simultaneous renewals are processed sequentially in a fixed atomic order;
- the physical system starts empty.

Set

$$r = \sqrt{c+1} - 1, \qquad \beta = 1 - \frac{r}{c}, \qquad \rho_e = \beta x_e.$$

Let $Y_e \sim \mathrm{Ber}(\rho_e)$ independently, set $Y = \sum_e Y_e$, and initialize

$$\widehat{S} \sim \mathcal{L}\big(\{e : Y_e = 1\} \mid Y \le c\big). \tag{10}$$

At a renewal of e:

1. end the preceding physical epoch and remove e from $\widehat{S}$;
2. if c other shadow coordinates remain, close the gate;
3. otherwise, open the gate with probability β;

4. insert e precisely when the gate opens and the fresh activation occurs.

For any configuration T of the other coordinates,

$$\Pr_{\mu}(e \in \widehat{S} \mid \widehat{S}_{-e} = T) = \begin{cases} \rho_e, & |T| \le c-1, \\ 0, & |T| = c. \end{cases}$$

Since $\rho_e = \beta x_e$, each update is exactly the coordinate Gibbs kernel for (10), so the law μ is invariant. The physical set remains contained in the shadow set pathwise.

For $x_e > 0$, the gate probability is

$$g_e = \beta \, \frac{\Pr(Y_{-e} \le c-1)}{\Pr(Y \le c)}. \tag{11}$$

Moreover,

$$\Pr(Y \le c) - \Pr(Y_{-e} \le c-1) = (1-\rho_e) \Pr(Y_{-e} = c) \le \Pr(Y = c).$$

Writing $b = \Pr(Y = c \mid Y \le c)$, we obtain

$$g_e \ge \beta(1-b). \tag{12}$$

If $b = 0$, this already gives $g_e \ge \beta$. Suppose therefore that $b > 0$, and let

$$\widetilde{Y} \sim \mathcal{L}(Y \mid Y \le c), \qquad J = c - \widetilde{Y}.$$

The Poisson-binomial mass function is log-concave: its generating polynomial has only real nonpositive roots, so Newton's inequalities apply. Truncation and reversal preserve log-concavity, so J has an increasing discrete hazard rate. Since $\Pr(J = 0) = b$,

$$\Pr(J \ge k) \le (1-b)^k, \qquad \mathbb{E}J \le \frac{1-b}{b}. \tag{13}$$

Conditioning on $Y \le c$ cannot increase the mean of Y, and therefore

$$\mathbb{E}J = c - \mathbb{E}[Y \mid Y \le c] \ge c - \mathbb{E}Y \ge c - \beta c = r.$$

Together with (13), this implies

$$b \le \frac{1}{r+1}.$$

Thus

$$g_e \ge \beta \, \frac{r}{r+1} = \frac{r}{r+2} = 1 - \frac{2}{\sqrt{c+1}+1}. \tag{14}$$

The previously omitted boundary cases are as follows.

- If $x_e = 0$, omit that chain or declare its gate always open; it never activates.
- If $b = 0$, use (12) directly, without dividing by b.
- If $c = 0$, the mass constraint forces every $x_e = 0$, so all gates may be declared open.
- Parallel updates are invalid: simultaneous renewals must be atomically ordered.

- A nonempty initial physical state is allowed only if it is coupled as a subset of the initial shadow state; the canonical empty initialization satisfies this automatically.

Initialization is efficient: rejection sampling from the independent variables Y_e succeeds with probability
$$\Pr(Y \le c) \ge 1 - \frac{\mathbb{E}Y}{c+1} \ge \frac{r+1}{c+1} = \frac{1}{\sqrt{c+1}},$$
so at most $O(\sqrt{c})$ trials are needed in expectation. If $c \ge |E|$, conditioning is unnecessary. Consequently, the expected running time is polynomial in $|E|$, and each renewal update takes constant time.

5 Why the recurring theorem still does not cover arbitrary intervals

Consider
$$c = 1, \qquad A_1 = 1, \qquad D_1 = \begin{cases} 1 & \text{with probability } 1/2, \\ 2 & \text{with probability } 1/2, \end{cases} \qquad A_2 \sim \mathrm{Ber}(1/2), \quad D_2 = 1.$$
The latent expected load is one at both times.

A single fixed chain cannot represent both opportunities, because on the branch $D_1 = 2$ it cannot renew at time 2. Assigning them to two fixed chains requires activation mass $1 + \frac{1}{2} > 1$. Splitting the first opportunity according to its realized duration is not causal, because its gate precedes D_1. This disproves the fixed-chain embedding; it does not disprove some more sophisticated configuration-valued construction.

Nor can scalar occupancy log-concavity supply the missing extension. With $c = 2$, eighteen permanent $\mathrm{Ber}(1/9)$ requests under accept-if-free produce
$$H = \min\{\mathrm{Bin}(18, 1/9), 2\},$$
with
$$\Pr(H = 0) \approx 0.120020, \qquad \Pr(H = 1) \approx 0.270046, \qquad \Pr(H = 2) \approx 0.609934,$$
and hence
$$\Pr(H = 1)^2 < \Pr(H = 0)\Pr(H = 2).$$
With heterogeneous returns, two histories of equal occupancy can also have completely different residual-duration configurations. A valid proof therefore needs a configuration-level invariant or a different causal policy.

The precise open problem is the following.

> *Given finite independent stochastic intervals with pointwise expected depth at most c, construct, in polynomial time, gates chosen before the current activation and duration such that the accepted depth is at most c pathwise and*
> $$\Pr(G_t = 1 \mid Z_t = z) \ge 1 - \frac{K}{\sqrt{c}}$$
> *for a universal constant K.*

Disposition of the rejection reasons

1. **Arbitrary-duration thinning theorem:** remains open; no $1 - K/\sqrt{c}$ construction or counterexample has been proved.

2. **Polynomial-time separation:** the erroneous computational-equivalence claim is removed. The separation is only information-theoretic; a polynomial-time construction remains open.

3. **Recurring-chain scope:** fully corrected in the theorem statement. The reduction from arbitrary intervals to that structure remains open and is not claimed.

4. **Boundary and specification omissions:** fully repaired; the cases $c = 0$ and $x_e = 0$, the zero boundary atom, initialization, schedule independence, simultaneous renewals, and efficient initialization are all treated explicitly.

5. **Maximum-independent-set argument:** removed. Under the source's assortment oracle it provides no obstruction.

Accordingly, the final result is substantive but partial: it gives the exact assortment-to-gate reduction, rigorous general upper and lower bounds, and a polynomial-time $1-O(c^{-1/2})$ theorem for fixed recurring chains, but not the requested arbitrary-duration competitive ratio.

References

[1] Yiding Feng, Rad Niazadeh, Amin Saberi. *Near-Optimal Bayesian Online Assortment of Reusable Resources*. Economics and Computation (EC), 2022. https://doi.org/10.1145/3490486.3538320